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An upwind numerical scheme for relativistic hydrodynamics with a general equation of state

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ABSTRACT

This paper describes a simple, second-order, upwind scheme for the equations of special relativistic hydrodynamics in multidimensions. A linear solver is used for all Riemann problems except those that contain strong rarefactions and, unlike previous algorithms, it is not restricted to an equation of state with a constant ratio of specific heats. Test calculations show that the scheme is accurate for flows containing extremely strong shocks and with Lorentz factors of several hundred. It has been combined with an adaptive hierarchical grid, which makes it possible to use high resolution in any number of dimensions.

Key words: equation of state – hydrodynamics – relativity – methods: numerical.

1 INTRODUCTION

Several recent papers (Eulderink & Mellema 1994; Font et al. 1994; Martí, Müller & Ibáñez 1994; Martí & Müller 1994b) have shown that modern upwind schemes can be used successfully to integrate the equations of relativistic hydrodynamics even when the Lorentz factor is large and the flow contains strong shocks. This is in contrast to non-upwind schemes, such as that described by Norman & Winkler (1986), which do not seem to be able to cope with Lorentz factors of much above 2 in more than one dimension.

However, these authors have all assumed a rather simple equation of state with constant ratio of specific heats, which is only valid if the particles are either ultrarelativistic or subrelativistic. This is a very severe restriction, which is only likely to be true for a limited class of problem. Furthermore, these codes can give unphysical results in some cases unless they use an exact Riemann solver. Although the Riemann problem can be solved exactly even with a realistic equation of state, this is much harder than in the classical case, and one would really like to avoid it if at all possible.

We have found that a very simple linear Riemann solver can deal with any shock problem and that it only fails if the flow contains strong rarefactions. However, Riemann problems whose solutions only contain rarefactions are reasonably easy to solve exactly, and since they are fairly uncommon, their cost is not particularly significant. When such a Riemann solver is used in the second-order scheme described by Falle (1991), the result is a robust algorithm which gives accurate results in all the cases we have tried so far. We have also combined this scheme with a hierarchical adaptive grid code, μCOBRA, which has been developed by Mantis Numerics Ltd for industrial applications.

2 RELATIVISTIC EQUATIONS

The energy and momentum equations of inviscid special relativistic hydrodynamics can be written (Taub 1978)

$$\frac{\partial T^{ij}}{\partial x^j}=0, \tag{2.1}$$

where $\boldsymbol{x}=(t, \boldsymbol{r})$ are the coordinates and $T^{ij}=wu^iu^j+g^{ij}p$ is the energy–momentum tensor. w is the enthalpy per unit volume in the fluid rest frame, p the pressure, $\boldsymbol{u}=(\Gamma, \Gamma\boldsymbol{v})$ the 4-velocity, and $\boldsymbol{v}$ the 3-velocity in units of c.

$$\Gamma=(1-v^2)^{-1/2} \tag{2.2}$$

is the Lorentz factor, and

$$g^{ij}=\begin{pmatrix}-1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix} \tag{2.3}$$

is the metric tensor in cartesian coordinates in Minkowski space. Here, and in what follows, we assume that latin indices i, j etc. range from 0 to 3 and greek indices α, β etc. from 1 to 3.

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We will suppose that the gas is composed of N species which are separately conserved. Particle conservation is then described by the equations

$$\frac{\partial D_I^j}{\partial x^j}=0, \tag{2.4}$$

where $D_I^j=n_I u^j$ is the 4-particle flux for species I and n_I is its density in the fluid rest frame. The total particle number density, n, is given by

$$n=\sum_{I=1}^{N} n_I. \tag{2.5}$$

Note that we have used capital I for the particle index to make it clear that it is not a tensor index.

These equations need to be supplemented by an equation of state connecting w, p and n_I ($I=1 \ldots N$). This equation of state is quite simple if the thermal velocities of the particles are either ultrarelativistic or subrelativistic in the fluid rest frame, but this only true in very special circumstances. More generally, for a perfect gas (Synge 1957)

$$w=\sum_{I=1}^{N} n_I m_I G(\xi_I), \tag{2.6}$$

$$p=\sum_{I=1}^{N} n_I m_I \xi_I^{-1}, \tag{2.7}$$

where m_I is the rest mass of species I and

$$\xi_I=\frac{m_I}{kT}, \tag{2.8}$$

$$G(\xi)=\frac{K_3(\xi)}{K_2(\xi)}=\frac{K_1(\xi)}{K_2(\xi)}+\frac{4}{\xi}. \tag{2.9}$$

Here, k is the Boltzmann constant, T the temperature and $K_\nu(\xi)$ are the modified Bessel functions. Equations (2.6)–(2.9) together constitute the required relation between w, p and the n_I.

The energy density (including the rest mass energy) is $\varepsilon=w-p$ and the sound speed, a, is given by

$$a^2=\left(\frac{\partial p}{\partial \varepsilon}\right)_s. \tag{2.10}$$

The ratio of specific heats for a relativistic ideal gas is

$$\gamma=\left\{\sum_{I=1}^{N} n_I G'(\xi_I)\xi_I^2\right\}\left\{\sum_{I=1}^{N} n_I[G'(\xi_I)\xi_I^2+1]\right\}^{-1}, \tag{2.11}$$

and one can use this to write (2.10) in the form

$$a^2=\frac{\gamma p}{w}. \tag{2.12}$$

Previous workers have defined a quantity, γ_*, by

$$w=\rho+\frac{\gamma_*}{(\gamma_*-1)}p. \tag{2.13}$$

Equations (2.6) and (2.7) then give

$$\frac{\gamma_*}{(\gamma_*-1)}=\left(\sum_{I=1}^{N} m_I n_I[G(\xi_I)-1]\right)\left(\sum_{I=1}^{N} m_I n_I/\xi_I\right)^{-1}. \tag{2.14}$$

Note that γ and γ_* are generally different functions of temperature, and are only constant and equal if the thermal energy of each species of particle remains either ultrarelativistic or non-relativistic. This will not, in general, be true for strong shocks or rarefactions in a flow with relativistic velocities. Fig. 1 shows γ and γ_* as functions of ξ for electron–positron and electron–proton gases. From these, it is clear that although a calculation with constant γ can capture the flavour of relativistic hydrodynamics, it will not generally be very accurate.

3 NUMERICAL METHOD

If we write equations (2.1) and (2.4) as evolution equations, we obtain

$$\frac{\partial}{\partial t}(wu^0u^0-p)+\frac{\partial}{\partial x^\alpha}(wu^0u^\alpha)=0$$

energy conservation, (3.1a)

$$\frac{\partial}{\partial t}(wu^0u^\beta)+\frac{\partial}{\partial x^\alpha}(wu^\alpha u^\beta+\delta^{\alpha\beta}p)=0$$

momentum conservation, (3.1b)

$$\frac{\partial}{\partial t}(\rho u^0)+\frac{\partial}{\partial x^\alpha}(\rho u^\alpha)=0$$

rest mass conservation, (3.1c)

$$\frac{\partial}{\partial t}(\rho c_I u^0)+\frac{\partial}{\partial x^\alpha}(\rho c_I u^\alpha)=0$$

conservation of species I particles ($I=1 \ldots N-1$). (3.1d)

Here $\delta^{\alpha\beta}$ is the Kronecker δ and c_I are the relative particle fractions

$$c_I=n_I\left(\sum_{I=1}^{N} n_I\right)^{-1}. \tag{3.2}$$

As these equations are a set of non-linear hyperbolic conservation laws, it seems sensible to use one of the modern upwind shock-capturing schemes that have been so successful in classical gas dynamics. This has already been done for the simple case of constant ratio of specific heats (Eulderink & Mellema 1994; Font et al. 1994; Martí et al. 1994; Martí & Müller 1994b), but, as we have already pointed out, this is a rather strong assumption which is only valid for a very limited class of problems.

3.1 The Riemann solver

As in all schemes of this type, we start by considering the one-dimensional Riemann problem for equations (3.1). In

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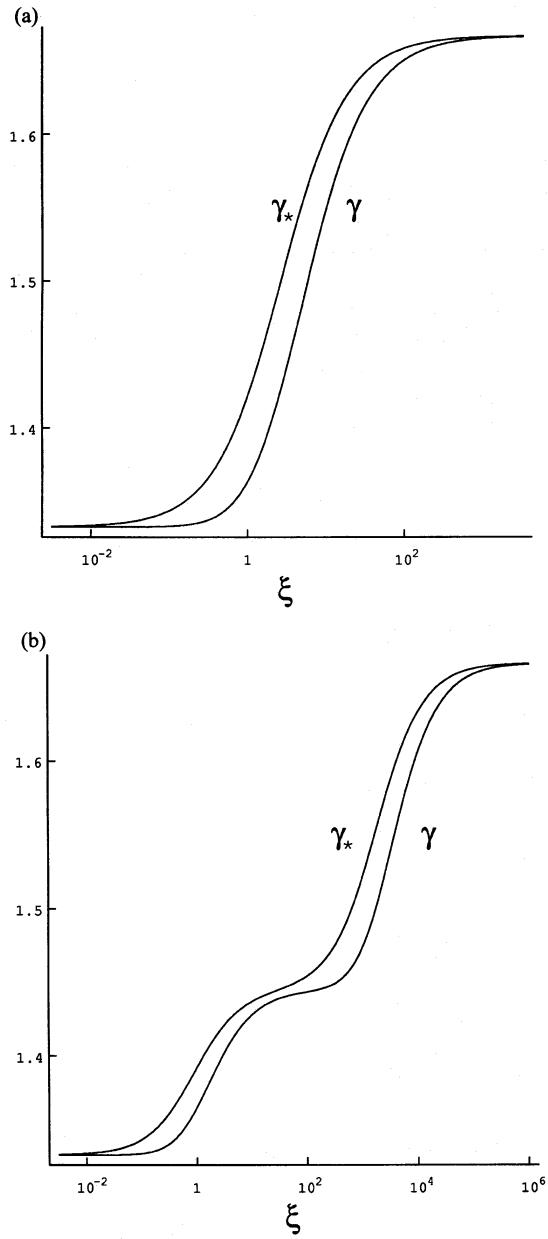


Figure 1. γ and γ_* as functions of inverse temperature ξ: (a) electron–positron gas; (b) electron–proton gas.

one dimension, these equations can be written in the form

$$\frac{\partial \boldsymbol{U}}{\partial t}+\frac{\partial \boldsymbol{F}}{\partial x}=0, \tag{3.3}$$

where

$$\boldsymbol{U}=[w\Gamma^2-p,\, w\Gamma^2\boldsymbol{v},\, \rho\Gamma,\, \rho\boldsymbol{c}\Gamma]^{\mathrm{T}} \tag{3.4}$$

is a vector of conserved quantities and

$$\boldsymbol{F}=[w\Gamma^2 v_x,\, w\Gamma^2 v_x^2+p,\, w\Gamma^2 v_y v_x,\, w\Gamma^2 v_z v_x,\, \rho\Gamma v_x,\, \rho\boldsymbol{c}\Gamma v_x]^{\mathrm{T}} \tag{3.5}$$

are the corresponding fluxes. Here, $\boldsymbol{c}$ is a vector of particle fractions

$$\boldsymbol{c}=(c_1,\ldots,c_{N-1})^{\mathrm{T}}$$

and the superscript 'T' denotes the transpose. Note that we have to allow for non-zero transverse velocities if we want to construct a multi-dimensional scheme.

The initial data in the Riemann problem is

$$\begin{aligned}\boldsymbol{U}&=\text{constant}=\boldsymbol{U}_{\mathrm{L}} \qquad \text{for } x\le 0,\\ &=\text{constant}=\boldsymbol{U}_{\mathrm{R}} \qquad \text{for } x>0.\end{aligned}$$

The exact solution to this problem is a similarity solution of the form

$$\boldsymbol{U}(x,t)=\boldsymbol{U}(\boldsymbol{U}_{\mathrm{L}},\boldsymbol{U}_{\mathrm{R}},x/t),$$

and we will call the (constant) state at $x=0$ the resolved state $\boldsymbol{U}^*(\boldsymbol{U}_{\mathrm{L}},\boldsymbol{U}_{\mathrm{R}})$.

Just as in classical gas dyamics, the solution to the Riemann problem consists of discontinuities (shocks or contact discontinuities) and centred rarefaction waves. Martí & Müller (1994a) have shown that if γ is constant and the transverse velocities are zero, the exact solution to the Riemann problem can be obtained by an iterative procedure similar to that used for classical gas dynamics (e.g. van Leer 1977). Although the exact resolved state can also be found in the general case, this is much more difficult, since there is neither an analytic solution for rarefaction waves nor an explicit expression for the shock adiabat. This is not too much of a problem in smooth regions where the difference between the left and right states is $0(h)$, where h is the mesh spacing. It is standard practice (e.g. Falle 1991; Toro 1993) to use a linearized Riemann solution for these and only invoke a non-linear solver inside shocks and sudden rarefactions.

It turns out that the algorithm described in Falle (1991) is able to handle strong shocks with a linear Riemann solver, and an iterative solution is only required at the centre of strong rarefaction fans, such as occur at the edge of highly underexpanded jets. This is a considerable advantage given the complexity of the relativistic Riemann problem. Furthermore, this particular scheme has been incorporated into a general-purpose hierarchical adaptive grid code (Falle & Giddings 1993), which means that very little modification is required to extend this code to relativistic hydrodynamics.

Obviously, all approximate Riemann solvers become identical as $\boldsymbol{U}_{\mathrm{L}}\to\boldsymbol{U}_{\mathrm{R}}$, but there is no reason why they should behave the same when the two states differ substantially. We have only tried two simple possibilities, but we find that, for relativistic hydrodynamics, there is actually quite a significant difference between them.

Riemann Solver A. Whether one uses an exact or approximate Riemann solver, the conserved variables are not particularly convenient and it is better to rewrite equation (3.3) in terms of other variables. There are many possibilities, but an obvious choice is

$$\boldsymbol{P}=(p,v_x,v_y,v_z,\rho,\boldsymbol{c}]^{\mathrm{T}}. \tag{3.6}$$

These satisfy the equations

$$\frac{\partial \boldsymbol{P}}{\partial t}+\mathbf{A}\frac{\partial \boldsymbol{P}}{\partial x}=0, \tag{3.7}$$

where the relevant properties of the matrix $\mathbf{A}$ and those for the y and z directions are given in Appendix A. In classical

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gas dynamics the primitive variables are given explicitly in terms of the conserved ones, but this is unfortunately not true in relativistic hydrodynamics even if the ratio of specific heats is constant. However, there is no difficulty in devising an efficient iterative procedure for determining the primitive variables from the conserved ones.

The most obvious linearization is that which is often used to generate the first guess for an iterative solver (van Leer 1977). We start by defining left and right matrices

$$\mathbf{A}_L=\mathbf{A}(\boldsymbol{U}_L), \qquad \mathbf{A}_R=\mathbf{A}(\boldsymbol{U}_R). \tag{3.8}$$

Let $\boldsymbol{R}_L^-$ be the right eigenvector of $\mathbf{A}_L$ associated with upstream moving sound waves and $\boldsymbol{R}_R^+$ be the right eigenvector of $\mathbf{A}_R$ for downstream moving sound waves. The linear approximation is then

$$\boldsymbol{P}_L^*=\boldsymbol{P}_L+b_L\boldsymbol{R}_L^-, \qquad \boldsymbol{P}_R^*=\boldsymbol{P}_R+b_R\boldsymbol{R}_R^+, \tag{3.9}$$

where $\boldsymbol{P}_L^*$ and $\boldsymbol{P}_R^*$ are the states on either side of the contact discontinuity. Since the pressure and x-component of the 3-velocity are continuous at the contact discontinuity, we must have

$$v_{xL}^*=v_{xR}^*, \qquad p_L^*=p_R^*, \tag{3.10}$$

Using equation (3.10) in (3.9), we get two linear equations, from which we can determine the wave strengths b_L, b_R and hence the linear approximation to the resolved state.

One might think that it would be better to use the 4-velocity rather than the 3-velocity, since for high Lorentz factors the 3-velocity is close to unity and is thus insensitive to changes in the flow. However, the matching condition at the contact discontinuity is that the normal components of the 3-velocity are equal and so this insensitivity cannot be removed merely by using the 4-velocity. Furthermore, the 4-velocity is less convenient if the transverse velocities are not zero, since the Lorentz factors can be different on either side of the contact and we no longer have explicit expressions for the resolved state. As it turns out, this particular Riemann solver works rather well, even for flows with large Lorentz factors.

This method can fail to give a positive pressure in the resolved state if the solution to the exact Riemann problem consists of two strong rarefactions. This difficulty can be overcome by numerically integrating along the rarefaction waves using

$$\frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}\mu_-}=\boldsymbol{R}^-(\boldsymbol{P}) \qquad \text{for upstream moving rarefactions,} \tag{3.11a}$$

$$\frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}\mu_+}=\boldsymbol{R}^+(\boldsymbol{P}) \qquad \text{for downstream moving rarefactions,} \tag{3.11b}$$

with the initial conditions

$$\boldsymbol{P}(0)=\boldsymbol{P}_L \qquad \text{for equation (3.11a)}$$
$$=\boldsymbol{P}_R \qquad \text{for equation (3.11b).}$$

Here μ_-, μ_+ are just parameters along the rarefaction curves in $\boldsymbol{P}$-space.

This gives us two solutions $\boldsymbol{P}^-(\mu_-)$, $\boldsymbol{P}^+(\mu_+)$ and the resolved state is determined by matching the x-component of the 3-velocity and the pressure, i.e. by finding μ_-^*, μ_+^* such that

$$v_x^-(\mu_-^*)=v_x^+(\mu_+^*), \qquad p^-(\mu_-^*)=p^+(\mu_+^*).$$

The states on the left and right of the contact discontinuity are then given by $\boldsymbol{P}^-(\mu_-^*)$ and $\boldsymbol{P}^+(\mu_+^*)$. If one of the rarefactions spans $x=0$, then that rarefaction must be transonic and the resolved state is that at the sonic point.

Riemann Solver B. If we define a mean matrix, instead of using different matrices on either side of the contact discontinuity, then it is possible to obtain explicit expressions for the resolved state even when the 4-velocity is used. In this case we work with the variables

$$\boldsymbol{Q}=[p, u_x, u_y, u_z, \rho, \boldsymbol{c}]^{\mathrm{T}}=[p, \Gamma v_x, \Gamma v_y, \Gamma v_z, \rho, \boldsymbol{c}]^{\mathrm{T}}, \tag{3.12}$$

which satisfy the equations

$$\frac{\partial \boldsymbol{Q}}{\partial t}+\mathbf{B}\frac{\partial \boldsymbol{Q}}{\partial x}=0. \tag{3.13}$$

The eigenvalues of $\mathbf{B}$ are obviously the same as those of the matrix $\mathbf{A}$, but the eigenvectors are different. The details are given in Appendix A.

We now define a mean matrix

$$\overline{\mathbf{B}}=\overline{\mathbf{B}}(\boldsymbol{P}_L, \boldsymbol{P}_R), \tag{3.14}$$

and solve the linear Riemann problem for the equation

$$\frac{\partial \boldsymbol{Q}}{\partial t}+\overline{\mathbf{B}}\frac{\partial \boldsymbol{Q}}{\partial x}=0. \tag{3.15}$$

The scheme will not be consistent unless

$$\overline{\mathbf{B}}(\boldsymbol{Q}, \boldsymbol{Q})=\mathbf{B}(\boldsymbol{Q}), \tag{3.16}$$

and $\overline{\mathbf{B}}$ must also have real eigenvalues and a complete set of linearly independent eigenvectors. These conditions are clearly not sufficient to determine $\overline{\mathbf{B}}$ uniquely and, as a consequence, there has been considerable debate about the respective merits of various choices.

Roe (1981) showed that, for classical gas dynamics, one can construct a $\overline{\mathbf{B}}$ which satisfies

$$\overline{\mathbf{B}}(\boldsymbol{Q}_L-\boldsymbol{Q}_R)=[\boldsymbol{F}(\boldsymbol{Q}_L)-\boldsymbol{F}(\boldsymbol{Q}_R)],$$

which ensures that the Riemann solver becomes exact when the solution only involves a single shock. However, it is not possible to find an explicit expression for such a matrix in the relativistic case with a realistic equation of state. In any case, there is no advantage in using such a matrix since it is not shocks but rarefactions which cause problems.

We therefore adopt the simplest possible form for $\overline{\mathbf{B}}$,

$$\overline{\mathbf{B}}(\boldsymbol{Q}_L, \boldsymbol{Q}_R)=\mathbf{B}[\tfrac{1}{2}(\boldsymbol{Q}_L+\boldsymbol{Q}_R)]. \tag{3.17}$$

Note that we have averaged the $\boldsymbol{Q}$ variables, because this does at least guarantee that the average is a physically possible state. This is not necessarily true for the average conserved variables in high Mach number flow, since this could lead to a negative average pressure. Equation (3.17) obviously satisfies (3.16), and since the average is a realistic state, $\overline{\mathbf{B}}$ does indeed have distinct real eigenvalues and linearly independent eigenvectors.

Once a mean matrix has been defined, the solution of the Riemann problem is straightforward (see e.g. Roe 1981). Let $\overline{\lambda}_J$, $\overline{\boldsymbol{R}}_J$, $\overline{\boldsymbol{L}}_J$ ($J=1, \ldots, N+5$) be the eigenvalues and left, right

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eigenvectors of $\bar{\mathbf{B}}$. If we normalize these so that

$$\bar{\boldsymbol{L}}_J \bar{\boldsymbol{R}}_K = \delta_{JK},$$

then we can write

$$\boldsymbol{Q}_R - \boldsymbol{Q}_L = \sum_{J=1}^{N+5} [\bar{\boldsymbol{L}}_J \cdot (\boldsymbol{Q}_R - \boldsymbol{Q}_L)] \bar{\boldsymbol{R}}_J.$$

The solution to the Riemann problem for equation (3.15) is then

$$\boldsymbol{Q}^* = \boldsymbol{Q}_L + \sum_{\lambda_J < 0} [\bar{\boldsymbol{L}}_J \cdot (\boldsymbol{Q}_R - \boldsymbol{Q}_L)] \bar{\boldsymbol{R}}_J$$

$$= \boldsymbol{Q}_R - \sum_{\lambda_J > 0} [\bar{\boldsymbol{L}}_J \cdot (\boldsymbol{Q}_R - \boldsymbol{Q}_L)] \bar{\boldsymbol{R}}_J.$$

This method also fails for strong rarefactions and these have to be dealt with in the same way as for method A.

3.2 Second-order scheme

In basing our scheme on the Riemann problem, we have ensured that it is upwind in the sense that it takes account of all the characteristics, not just the streamlines. We also need to make it conservative since this is necessary to guarantee the correct jump conditions at shocks.

In cartesian coordinates, strict conservation is possible, but it is not in axisymmetry because we have to work with the radial momentum, which is not a conserved quantity. However, by using a finite-volume formulation, it is possible to construct an algorithm which becomes conservative at shocks as the mesh spacing tends to zero. Of course we cannot actually go to the limit of zero mesh spacing, but this only causes problems near the axis of symmetry and these can be minimized by ensuring that the flow is well resolved there.

We will describe the scheme for cylindrical polars with axisymmetry, since this illustrates the basic ideas and indicates how it can be extended to other geometries. In these coordinates, equation (2.1) can be written in the form

$$\frac{\partial \boldsymbol{U}}{\partial t} + \frac{1}{r}\frac{\partial r\boldsymbol{F}}{\partial r} + \frac{\partial \boldsymbol{G}}{\partial z} = \boldsymbol{S}, \tag{3.18}$$

where

$$\boldsymbol{U} = [w\Gamma^2 - p,\ w\Gamma^2 v_r,\ w\Gamma^2 v_z,\ \rho\Gamma,\ \rho c\Gamma]^{\mathrm{T}}$$

is the vector of conserved quantities and

$$\boldsymbol{F} = [w\Gamma v_r,\ w\Gamma^2 v_r^2 + p,\ w\Gamma^2 v_r v_z,\ \rho\Gamma v_r,\ \rho c\Gamma v_r]^{\mathrm{T}},$$

$$\boldsymbol{G} = [w\Gamma v_z,\ w\Gamma^2 v_r v_z,\ w\Gamma^2 v_z^2 + p,\ \rho\Gamma v_z,\ \rho c\Gamma v_z]^{\mathrm{T}},$$

are the fluxes in the r and z directions. The source term is given by

$$\boldsymbol{S} = \left[0, \frac{p}{r}, 0, 0, \boldsymbol{0}\right] \tag{3.19}$$

and accounts for the fact that the radial momentum is not conserved.

To construct a scheme for equation (3.18), we divide the computational domain into cells with mesh spacing h such that the i, j cell occupies the region

$$(i-1)h \le r \le ih, \qquad (j-1)h \le z \le jh,$$

whose volume is

$$V_{ij} = \pi(2i-1)h^3.$$

Now suppose that we know the solution at some time $t = t_k$ and we want to calculate that at a later time t_{k+1}. We integrate (3.18) over the i, j cell and from $t = t_k$ to $t = t_{k+1}$ to get

$$\frac{\boldsymbol{U}_{ijk+1} - \boldsymbol{U}_{ijk}}{t_{k+1} - t_k} + \frac{2}{(2i-1)h}(i\boldsymbol{F}_{i+1/2jk+1/2} - (i-1)\boldsymbol{F}_{i-1/2jk+1/2}$$

$$+ \frac{1}{h}(\boldsymbol{G}_{ij+1/2k+1/2} - \boldsymbol{G}_{ij-1/2k+1/2}) = \boldsymbol{S}_{ijk+1/2}. \tag{3.20}$$

Here

$$\boldsymbol{U}_{ijk} = \frac{2\pi}{V_{ij}} \int_{(j-1)h}^{jh} \int_{(i-1)h}^{ih} \boldsymbol{U}(r, z, t_k)\, r\, \mathrm{d}r\, \mathrm{d}z,$$

is the mean value of $\boldsymbol{U}$ in the i, j cell at time t_k,

$$\boldsymbol{F}_{i+1/2jk+1/2} = \frac{1}{(t_{k+1} - t_k)h} \int_{t_k}^{t_{k+1}} \int_{(j-1)h}^{jh} \boldsymbol{F}(ih, z, t)\, \mathrm{d}z\, \mathrm{d}t,$$

$$\boldsymbol{G}_{ij+1/2k+1/2} = \frac{2}{(t_{k+1} - t_k)(2i-1)h} \int_{t_k}^{t_{k+1}} \int_{(i-1)h}^{ih} \boldsymbol{G}(r, jh, t)\, r\, \mathrm{d}r\, \mathrm{d}t,$$

are the fluxes averaged over time and the cell interfaces and

$$\boldsymbol{S}_{ijk+1/2} = \frac{2\pi}{(t_{k+1} - t_k) V_{ij}} \int_{t_k}^{t_{k+1}} \int_{(j-1)h}^{jh} \int_{(i-1)h}^{ih} \boldsymbol{S}(r, z, t)\, r\, \mathrm{d}r\, \mathrm{d}z\, \mathrm{d}t, \tag{3.21}$$

is the source term averaged over time and the volume of the cell.

Equation (3.20) is exact and forms the basis of all conservative schemes for hyperbolic conservation laws. It is the way in which the quantities with half-integral suffices are approximated that distinguishes one such scheme from another.

In a first-order Godunov-type scheme the approximations to the fluxes and source term are derived by assuming that the solution is uniform within each cell and constant over a time-step. The fluxes are determined by ignoring the source terms and solving a one-dimensional Riemann problem at the cell interfaces.

As before, we denote the resolved state by $\boldsymbol{U}^*(\boldsymbol{U}_L, \boldsymbol{U}_R)$. For a first-order scheme, the fluxes are then

$$\boldsymbol{F}_{i+1/2jk+1/2} = \boldsymbol{F}[\boldsymbol{U}^*(\boldsymbol{U}_{ijk}, \boldsymbol{U}_{i+1jk})],$$

$$\boldsymbol{G}_{ij+1/2k+1/2} = \boldsymbol{G}[\boldsymbol{U}_*(\boldsymbol{U}_{ijk}, \boldsymbol{U}_{ij+1k})],$$

i.e. the left and right states are those in the cells on either side of the interface and the flux is that in the corresponding resolved state.

The source term has to be treated with some care. It is no good assuming that $\boldsymbol{S}$ is uniform in each cell since it depends upon the radial coordinate. Instead, we have to evaluate $\boldsymbol{S}$ by setting the pressure constant and then integrating over the cell. The result for the non-zero component of $\boldsymbol{S}$ is

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$$\left(\frac{p}{r}\right)_{ijk+1/2} = \frac{2\pi}{(t_{k+1}-t_k)V_{ij}} p_{ijk} \int_{t_k}^{t_{k+1}} \int_{(j-1)h}^{jh} \int_{(i-1)h}^{ih} \mathrm{d}r\,\mathrm{d}z\,\mathrm{d}t$$

$$= \frac{2p_{ijk}}{(2i-1)h}.$$

This form of $\boldsymbol{S}$ ensures that a state with uniform pressure and zero velocity is an exact steady solution of the difference equations.

The above is a straightforward extension of Godunov's scheme to axisymmetry. To make this second-order in time and space we have to introduce some structure inside the cells and allow for the variation in $\boldsymbol{U}$ over the time-step. Second-order time accuracy can be obtained by using the first-order scheme to obtain an intermediate solution $\boldsymbol{U}_{ijk+1/2}$ at the half-time $t=(t_{k+1}+t_k)/2$, which is then used to determine the second-order fluxes.

Second-order space accuracy requires a knowledge of the gradients within each cell, but we cannot just use central differences for this since Godunov's theorem (Godunov 1959) warns us that such a scheme will not be monotonic in the neighbourhood of shocks and tangential discontinuities. Instead, we have to use a non-linear switch which has the effect of reducing the order of the scheme in such regions. Broadly speaking there are two approaches to this. The first is flux-corrected transport, which uses a flux limiter to modify the second-order flux so that it does not cause oscillations in regions where the second derivatives are large (Boris & Book 1973). Although this is a sound idea, it is very complicated in more than one dimension, and has a number of unpleasant side effects. The alternative is to apply a non-linear averaging function to the gradients which has the effect of taking the smallest of the two gradients on either side of the cell (van Leer 1977).

We will adopt van Leer's approach, except that it is better to work with the variables defined by equation (3.12). In axisymmetry, these reduce to

$$\boldsymbol{Q}=[p, u_r, u_z, \rho, \boldsymbol{c}].$$

The gradients in these variables are calculated from

$$\left(\frac{\partial \boldsymbol{Q}}{\partial r}\right)_{ijk+1/2} = av\left(\frac{\boldsymbol{Q}_{ijk+1/2}-\boldsymbol{Q}_{i-1jk+1/2}}{r_g(i)-r_g(i-1)}, \frac{\boldsymbol{Q}_{i+1jk+1/2}-\boldsymbol{Q}_{ijk+1/2}}{r_g(i+1)-r_g(i)}\right),$$

$$\left(\frac{\partial \boldsymbol{Q}}{\partial z}\right)_{ijk+1/2} = av\left(\frac{\boldsymbol{Q}_{ijk+1/2}-\boldsymbol{Q}_{ij-1k+1/2}}{h}, \frac{\boldsymbol{Q}_{ij+1k+1/2}-\boldsymbol{Q}_{ijk+1/2}}{h}\right),$$

where $av(a, b)$ is the averaging function and $r_g(i)$ is the radial coordinate of centre of gravity of the cells with r-index i,

$$r_g(i) = \frac{2}{3}\frac{(3i^2-3i+1)}{(2i-1)}.$$

The averaging function has to have the following properties

$$av(a, b) \to \tfrac{1}{2}(a+b) \text{ as } a \to b,$$

$$= 0 \qquad \text{if } ab < 0,$$

$$\to a \qquad \text{as } |b|/|a| \to \infty,$$

$$\to b \qquad \text{as } |a|/|b| \to \infty.$$

There are obviously an infinite number of functions with these properties, but our attitude is that the simplest is the best. We therefore use

$$av(a, b) = \frac{(a^2 b + ab^2)}{(a^2+b^2)} \qquad \text{if } a^2+b^2>0 \text{ and } ab>0,$$

$$= 0 \qquad \text{if } ab \le 0,$$

which clearly has the desired behaviour.

Once we have obtained suitable gradients in each cell, we can set up the left and right states for the Riemann problems. For the radial fluxes we have

$$\boldsymbol{Q}_L = Q_{ijk+1/2} + [ih - r_g(i)]\left(\frac{\partial \boldsymbol{Q}}{\partial r}\right)_{ijk+1/2},$$

$$\boldsymbol{Q}_R = Q_{i+1jk+1/2} + [ih - r_g(i+1)]\left(\frac{\partial \boldsymbol{Q}}{\partial r}\right)_{i+1jk+1/2},$$

$$\boldsymbol{F}_{i+1/2jk+1/2} = \boldsymbol{F}\{\boldsymbol{U}^*[\boldsymbol{U}(\boldsymbol{Q}_L), \boldsymbol{U}(\boldsymbol{Q}_R)]\},$$

and similarly for the axial fluxes. Note that although we use the $\boldsymbol{Q}$ variables for interpolation for both Riemann solvers, these have to be converted to $\boldsymbol{P}$ variables before we can use Riemann solver A. This conversion is trivial, since

$$\Gamma^2 = 1 + (u_r^2 + u_z^2)$$

from the definition of the 4-velocity.

Having dealt with the fluxes, we must now derive an expression for the source term. All we have to do is to use our linear pressure distribution to evaluate the integral in equation (3.21). The result is

$$\left(\frac{p}{r}\right)_{ijk+1/2} =$$

$$\frac{2}{(2i-1)h}\left\{p_{ijk+1/2} + \left[h\left(i-\frac{1}{2}\right) - r_g(i)\right]\left(\frac{\partial p}{\partial r}\right)_{ijk+1/2}\right\}.$$

Once we have calculated the second-order fluxes and source term in this way, we can use them in equation (3.20) to advance the solution through the complete time-step. The resulting explicit upwind scheme is as conservative as it is possible to be in axisymmetry, and is second-order in smooth regions. Being explicit, it is subject to the usual Courant stability condition

$$t_{k+1} - t_k \le \min_{ij}\left(\frac{h}{\lambda_{ijk}}\right),$$

where λ_{ijk} is the maximum wavespeed in the i, j cell at time $t=t_k$.

This scheme is simply the relativistic version of the one described in Falle (1991) for classical gas dynamics. It has been applied, with some success, to a number of classical problems (e.g. Falle & Wilson 1988; Falle 1991, 1994; Falle & Raga 1993, 1995). Although it is much simpler than PPM, it is much safer in that it is not antidiffusive and therefore does not tend to generate unnaturally sharp features.

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3.3 Artificial viscosity

Although upwind schemes generally perform extremely well, they suffer from a number of problems if they rely entirely on the dissipation generated by the truncation error, since this becomes very anisotropic if the flow is nearly aligned with the grid. Not only can this lead to unphysical distortions of shock fronts, but it also allows the growth of a non-linear instability of plane shocks (Quirk 1992). All of these faults can be remedied by adding some extra dissipation to the Riemann solver. This also minimizes the long-wavelength entropy errors behind slowly moving shocks (Roberts 1988).

We have to add artificial viscosity at shocks, but in doing this we must be careful not to degrade the performance of the scheme in smooth regions. We therefore use the left and right states in the Riemann problem to define a viscous stress

$$\tau^{\alpha\beta}=\eta w^* a^*(u^\alpha_{\rm L}-u^\alpha_{\rm R}), \qquad (3.22)$$

where w^*, a^* are the enthalpy and sound speed in the resolved state, $u^\alpha_{\rm L}$, $u^\alpha_{\rm R}$ are the 4-velocities in the left and right states, and η is a constant parameter (typically $\eta=0.1$–0.2). The superfix β refers to the direction associated with the Riemann problem. Note that $\tau^{\alpha\beta}$ is not Lorentz-invariant, but then neither is any Eulerian scheme for the relativistic equations, nor any such schemes Galilean-invariant for classical gas dynamics.

In regions where the scheme becomes first-order, we have

$$(u^\alpha_{\rm L}-u^\alpha_{\rm R})\simeq h\frac{\partial u^\alpha}{\partial x^\beta}$$

in the β direction, which means that (3.22) corresponds to a kinematic viscosity, ν, of the form

$$\nu=\eta a h.$$

Table 1. Parameters used in the numerical calculations.

Case 1 (One Dimensional)	
Domain and Grid	$0 \le x \le 1$, uniform, 400 cells.
Boundary Conditions	Free flow at $x = 0$ and $x = 1$.
Initial State	$\rho = 1$, $p = 10^3$, $u = 0$ for $x \le 0.5$,
	$\rho = 1$, $p = 10^{-2}$, $u = 0$ for $x > 0.5$.
Case 1 (Two Dimensional)	
Domain and Grid	$0 \le x \le 1$, $0 \le y \le 1$, adaptive with 5 levels, coarsest grid 30x30, finest grid 480x480.
Boundary Conditions	Free flow on all boundaries.
Initial State	$\rho = 1$, $p = 10^3$, $u = 0$ for $y + \sqrt{3}x \le \frac{1}{2}(1 + \sqrt{3})$,
	$\rho = 1$, $p = 10^{-2}$, $u = 0$ for $y + \sqrt{3}x > \frac{1}{2}(1 + \sqrt{3})$.
Case 2	
Domain and Grid	$0 \le x \le 1$, uniform, 400 cells.
Boundary Conditions	Inflow at $x = 0$, symmetry at $x = 1$.
Initial State	$\rho = 1$, $p = 1$, $u = -2.236\times10^2$.
Case 3 (Spherically Symmetric)	
Domain and Grid	$0 \le r \le 4$, uniform, 100 cells.
Boundary Conditions	Free flow at $r = 4$.
Initial State	$\rho = 10$, $p = 10^3$, $u = 0$ for $r \le 0.5$,
	$\rho = 10$, $p = 3\times10^{-3}$, $u = 0$ for $r > 0.5$.
Case 3 (Axisymmetric)	
Domain and Grid	$0 \le r \le 4$, $0 \le z \le 4$, uniform, 100 cells in r, z.
Boundary Conditions	Symmetry at $z = 0$, free flow at $r = 4$ and $z = 4$.
Initial State	$\rho = 10$, $p = 10^3$, $u = 0$ for $\sqrt{(r^2 + z^2)} \le 0.5$,
	$\rho = 10$, $p = 3\times10^{-3}$, $u = 0$ for $\sqrt{(r^2 + z^2)} > 0.5$.
Case 4 (Axisymmetric Jet)	
Domain and Grid	$0 \le r \le 8$, $0 \le z \le 10$, adaptive with 6 levels, coarsest grid 13x16, finest grid 416x512.
Initial Jet	$\rho = 1$, $p = 1$, $u^r = 0$, $u^z = 7$, initial radius 0.3.
Boundary Conditions	Inflow in the jet, symmetry at $z = 0$, free flow at $r = 8$ and $z = 10$.
Initial State	$\rho = 10$, $p = 1$, $u = 0$.

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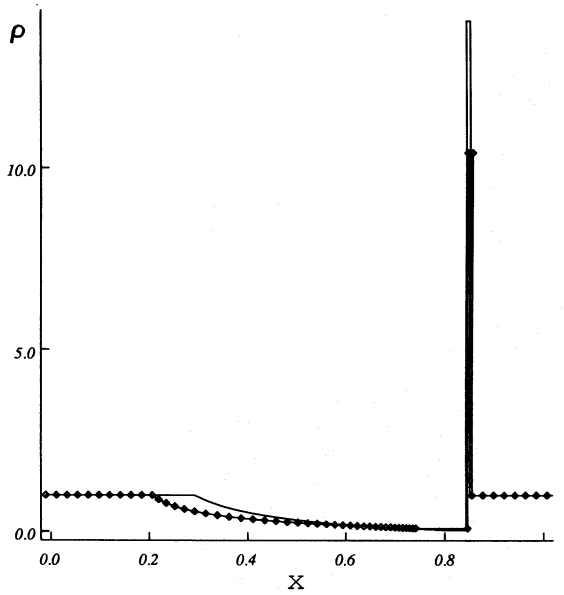


Figure 2. Density in the exact solution for case 1 for $\gamma = \text{constant} = 5/3$ (markers) and for the general equation of state for an electron–proton gas (line).

However, in smooth regions

$$(u^\alpha_{\rm L} - u^\alpha_{\rm R}) \simeq h^2 \frac{\partial^2 u^\alpha}{\partial x^\beta \, \partial x^\beta},$$

and the viscosity is roughly

$$\nu \simeq \eta a h^2 \left(\frac{\partial^2 u^\alpha}{\partial x^\beta \, \partial x^\beta} \right) \left(\frac{\partial u^\alpha}{\partial x^\beta} \right)^{-1}$$

Since this is second-order, it does not reduce the order of the scheme in such regions.

The role of artificial viscosity in our scheme is quite different from that in Norman & Winkler (1986). Because their scheme was neither upwind nor conservative, they were forced to use a very large viscosity to smear shocks over many mesh points. This meant that the viscous stresses were so large that they had to be included in the total energy. Our viscous stresses have negligible energy density, even in shocks, and their only purpose is to remove some minor faults that only occur in special circumstances.

3.4 Adaptive grid

One of the main difficulties with relativistic hydrodynamics is that the relation between mass and energy density implies that there is no limit to the compression ratio across a shock as the Lorentz factor tends to infinity. Such shocks therefore behave somewhat like isothermal shocks in ordinary gas dynamics, and usually produce very thin dense regions, which are very hard for any numerical scheme to resolve adequately. This has already been pointed out by Norman & Winkler (1986) and they used an adaptive grid to overcome this problem. However, their algorithm required an implicit scheme and would be very difficult to implement in more than one dimension.

We have implemented the above numerical scheme in a hierarchical adaptive grid code, μCOBRA, which has been developed by Mantis Numerics Ltd for industrial applications (Falle & Giddings 1993). μCOBRA uses a hierarchy of grids G^0, $G^1 \ldots G^N$, such that the mesh spacing on G^n is $h/2^n$, and the spacing on the coarsest grid is h. Grids G^0 and G^1 cover the whole domain, but the finer grids need only exist in regions which require high resolution. The grid hierarchy is used to generate an estimate of the truncation error by comparing solutions on grids with different mesh spacings, and the grid is refined if this error exceeds a given tolerance. Refinement also occurs in time, so that if the time-step on G^0 is Δt, then it is $\Delta t/2^n$ for G^n. The code is extremely general and works equally well in one, two and three dimensions. It has already been used to model both stellar and extragalactic jets (Falle & Raga 1993; Falle 1994; Falle & Raga 1995).

4 NUMERICAL RESULTS

In order to evaluate the scheme described in the previous section, we have to try it out on a set of test problems for which we know the exact solution and which bears some relation to those that occur in practice. This is obviously even harder for relativistic gas dynamics than it is in the classical case, but we do at least have exact solutions for one-dimensional shock-tube problems. These have the advantage that they contain shocks, contact discontinuities and rarefaction waves, and their oblique versions can be used to test the scheme in two dimensions.

Case 1. We start with a rather mild case which has already been considered by several authors (Norman & Winkler 1986; Martí & Müller 1994b). The parameters of the problem are given in Table 1 and are the same as those for problem 2 in Martí & Müller (1994b). This is not a particularly demanding problem, since the maximum Lorentz factor of the flow is only 3.57, which is small enough for a non-conservative code to cope with (Norman & Winkler 1986). As previous calculations have assumed a fixed value of γ ($\gamma = 5/3$), we must obviously do the same if we are to compare our results with theirs. This is not appropriate for this problem, however, since $\gamma = 4/3$ in the left state, $\gamma = 1.39$ behind the shock and $\gamma = 1.45$ in the right state. Fig. 2 shows that the correct equation of state for an electron–proton gas gives a solution which differs significantly from the one with $\gamma = 5/3$ everywhere.

Figs 3 and 4 show the exact and numerical solutions for Riemann solvers A and B with $\gamma = 5/3$. Riemann solver B is unable to cope with the initial discontinuity, so we had to start the calculation by using Riemann solver A for 10 time-steps, and only then could we switch to Riemann solver B. Apart from this, there is no detectable difference between them. Both of them agree extremely well with the analytic solution everywhere except in the high-density region behind the shock, where numerical diffusion reduces the peak density to 54 per cent of the theoretical value. At the same resolution, PPM gives 69 per cent of the theoretical value (Martí & Müller 1994b), which is hardly surprising since PPM is antidiffusive and should therefore do particularly well on this kind of problem.

It is possible to use the adaptive grid described in the previous section to resolve such features. Fig. 3(d) shows the

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density for an adaptive calculation with seven levels, giving a finest grid that is eight times finer than the uniform grid used in Figs 3(a)–(c). The peak density is now 98 per cent of the theoretical value, which means that the convergence rate is somewhat better than first order. Note that, although the adaptive calculation has much higher resolution in the dense shell, the lower resolution in the smooth part of the flow means that it is significantly faster than the uniform grid calculation.

Fig. 5 shows that our scheme also works for the exact equation of state, although the higher post-shock density means that numerical diffusion is more of the problem. As we have already pointed out, this difficulty always occurs for strong ultrarelativistic shocks and the only viable option is to use an adaptive grid. We have used a five-level adaptive grid on the oblique version of this problem, since this would be quite an expensive calculation on a uniform grid. Fig. 6 shows that a calculation with the initial discontinuity oriented at 30° to the x-axis gives much the same results as in one dimension.

Case 2. A collision between two equal gas streams is in some ways a more demanding test of a shock-capturing code since the gas is stationary behind the shocks, so there is no heat diffusion to iron out the entropy errors. This always leads to an entropy deficiency at the collision point and can

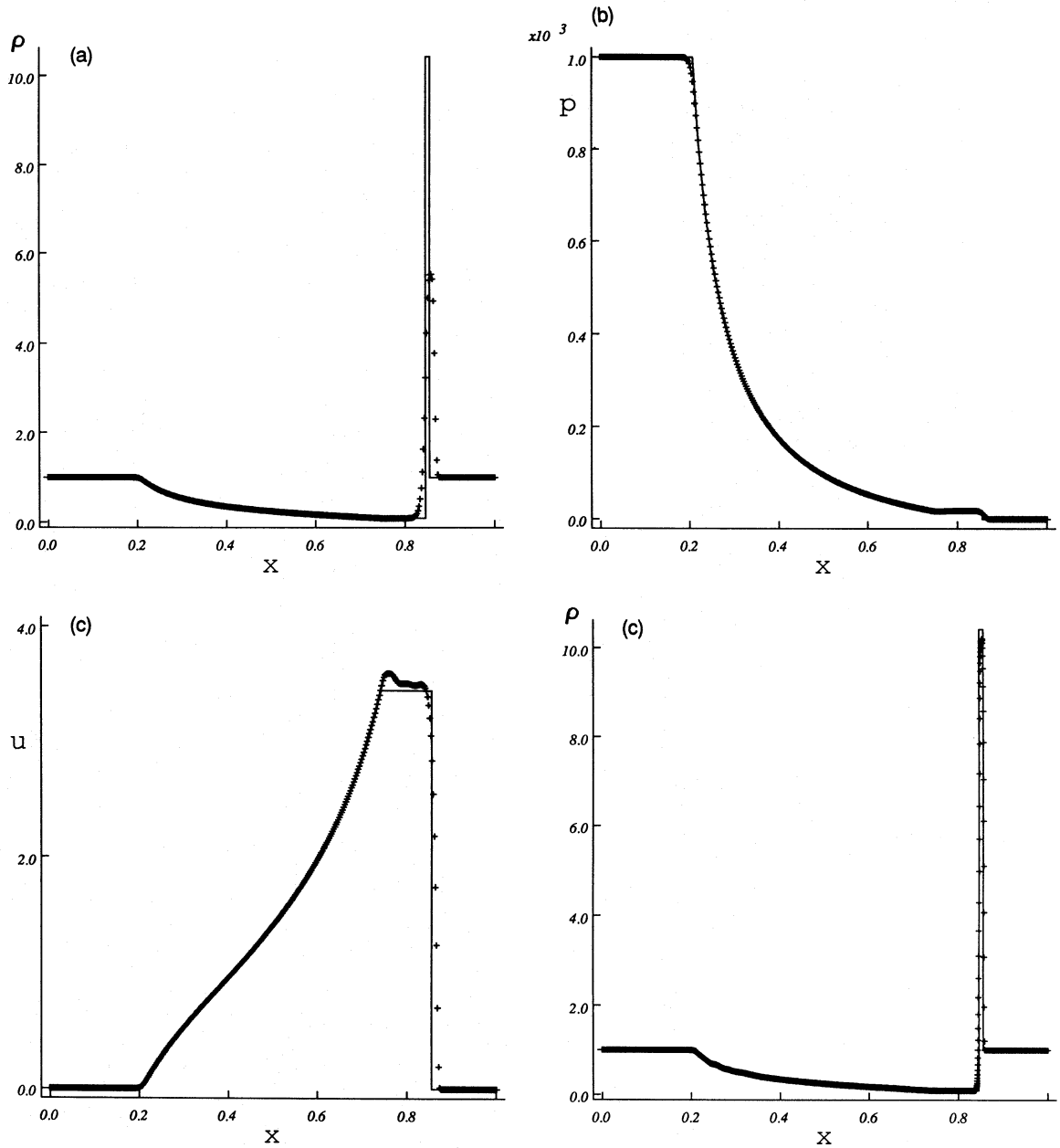


Figure 3. Exact solution (line) and numerical calculation (markers) for case 1 with $\gamma = 5/3$ at $t = 0.36$ (Riemann solver A): (a) density; (b) pressure; (c) x-component of 4-velocity; (d) density from seven-level adaptive calculation with a finest grid spacing corresponding to 3200 cells.

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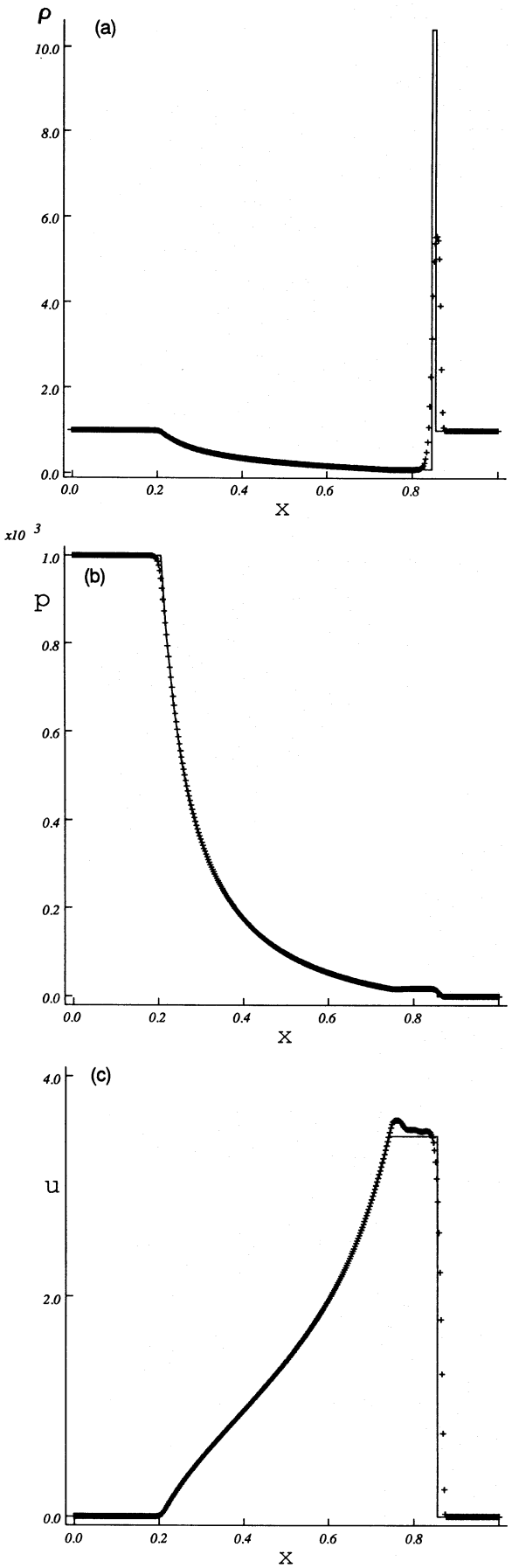


Figure 4. Exact solution (line) and numerical calculation (markers) for case 1 with $\gamma = 5/3$ at $t = 0.36$ (Riemann solver B): (a) density; (b) pressure; (c) x-component of 4-velocity.

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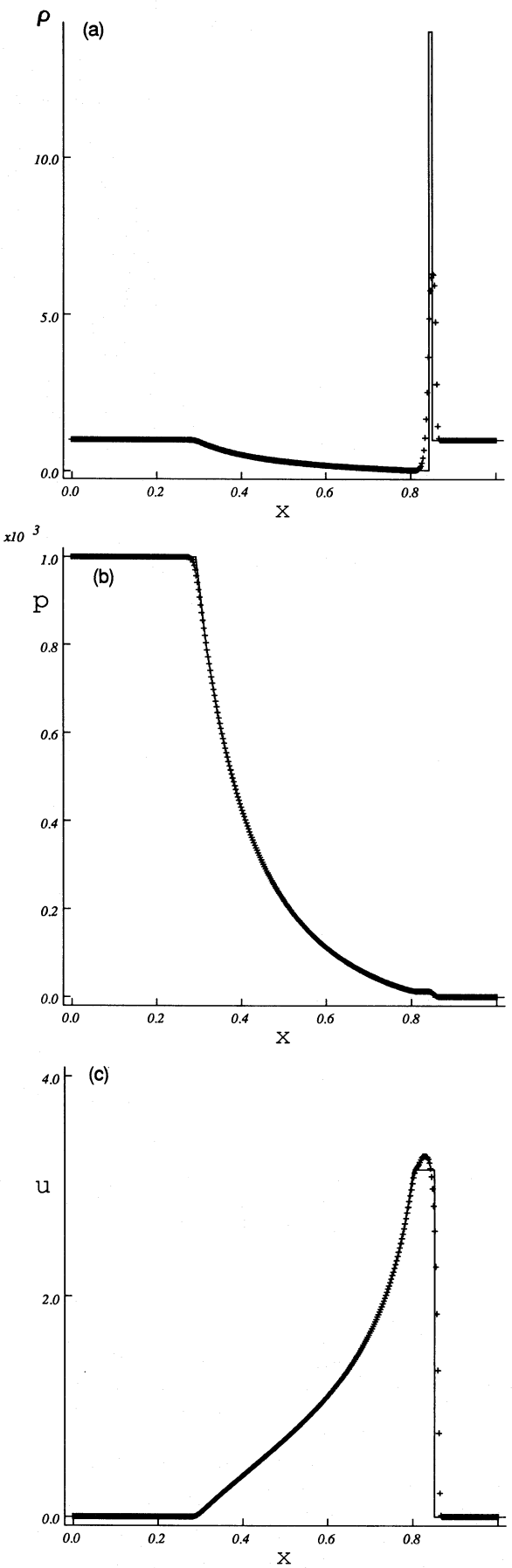


Figure 5. Exact solution (line) numerical calculation (markers) for case 1 with the general equation of state for an electron–proton gas at $t = 0.36$ (Riemann solver A): (a) density; (b) pressure; (c) x-component of 4-velocity.

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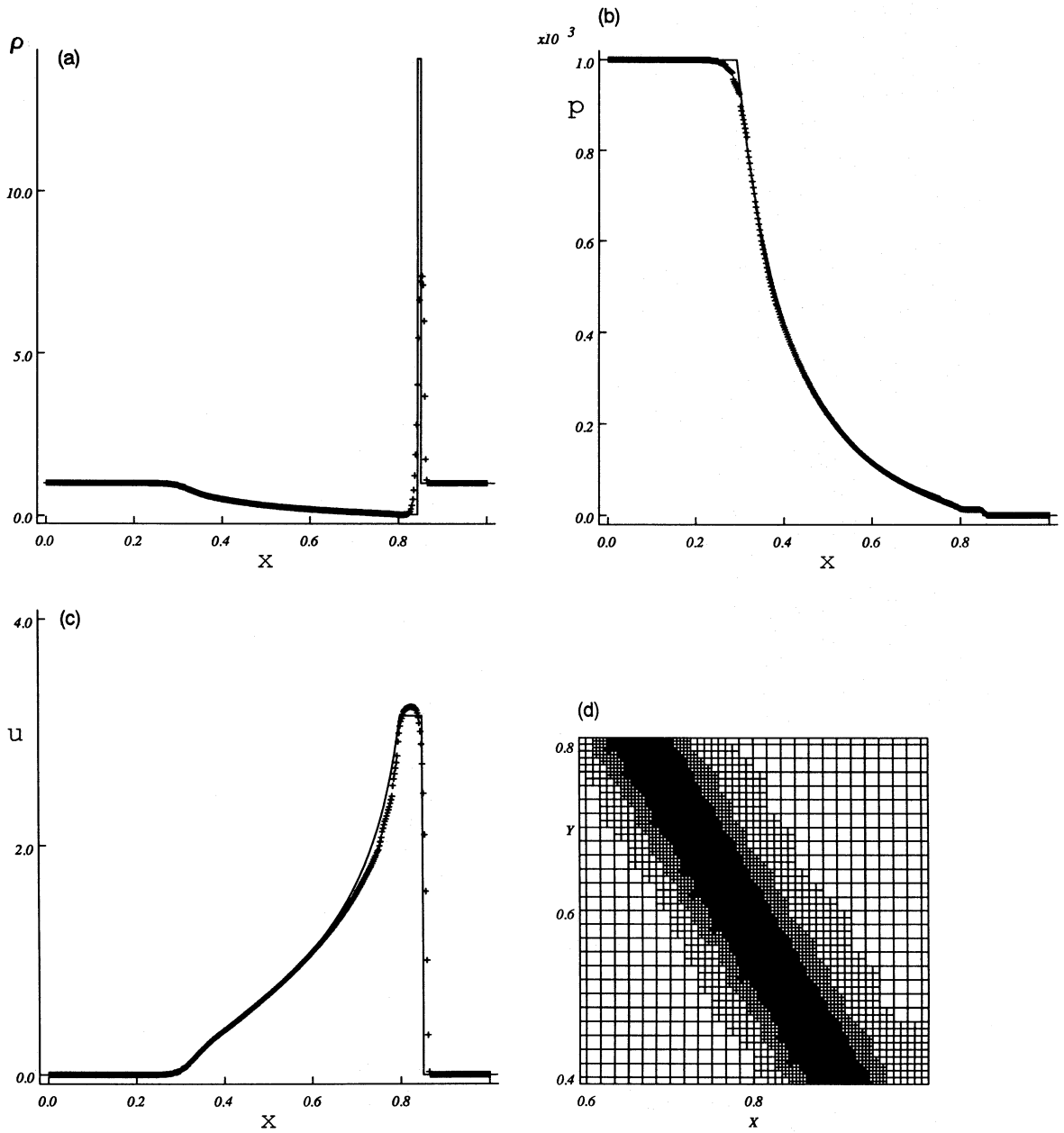


Figure 6. A cut at $y = 0.5$ showing the exact solution (line) and numerical calculation (markers) for the oblique case 1 with the general equation of state for an electron–proton gas (Riemann solver A). $t = 0.312$ which gives the same width as in Fig. 5: (a) density; (b) pressure; (c) space component of 4-velocity; (d) grids G^1 to G^4.

introduce an oscillatory entropy error behind a shock which is moving slowly in the sense that it takes many time-steps for it to traverse one grid point (Roberts 1988). Although the shock considered by Martí & Müller (1994b) is slowly moving in this sense, for some reason it does not generate oscillatory entropy errors.

Riemann solver B fails completely on this problem, even if we start the calculation with Riemann solver A. However, Fig. 7 shows that Riemann solver A gives just as good results as the PPM calculation in Martí & Müller (1994b). This is to be expected, since even a first-order upwind scheme would capture such a slowly moving shock very sharply. Note that we used $\gamma = 4/3$ in this calculation, but it would make no difference if we used the exact equation of state, since the gas behind the shock is ultrarelativistic.

Case 3. The similarity solution for a strong spherical explosion provides a useful test problem in classical gas dynamics (Arthur & Falle 1991), but in the relativistic case it is impossible for an explosion to release a finite amount of energy and zero mass. We can nevertheless test our axisymmetric code by comparing it with a one-dimensional spherically symmetric calculation. Fig. 8 shows the results of such a comparison, and we can see that the one- and two-dimensional calculations give almost identical results. This, together with the results for the oblique shock-tube problem,

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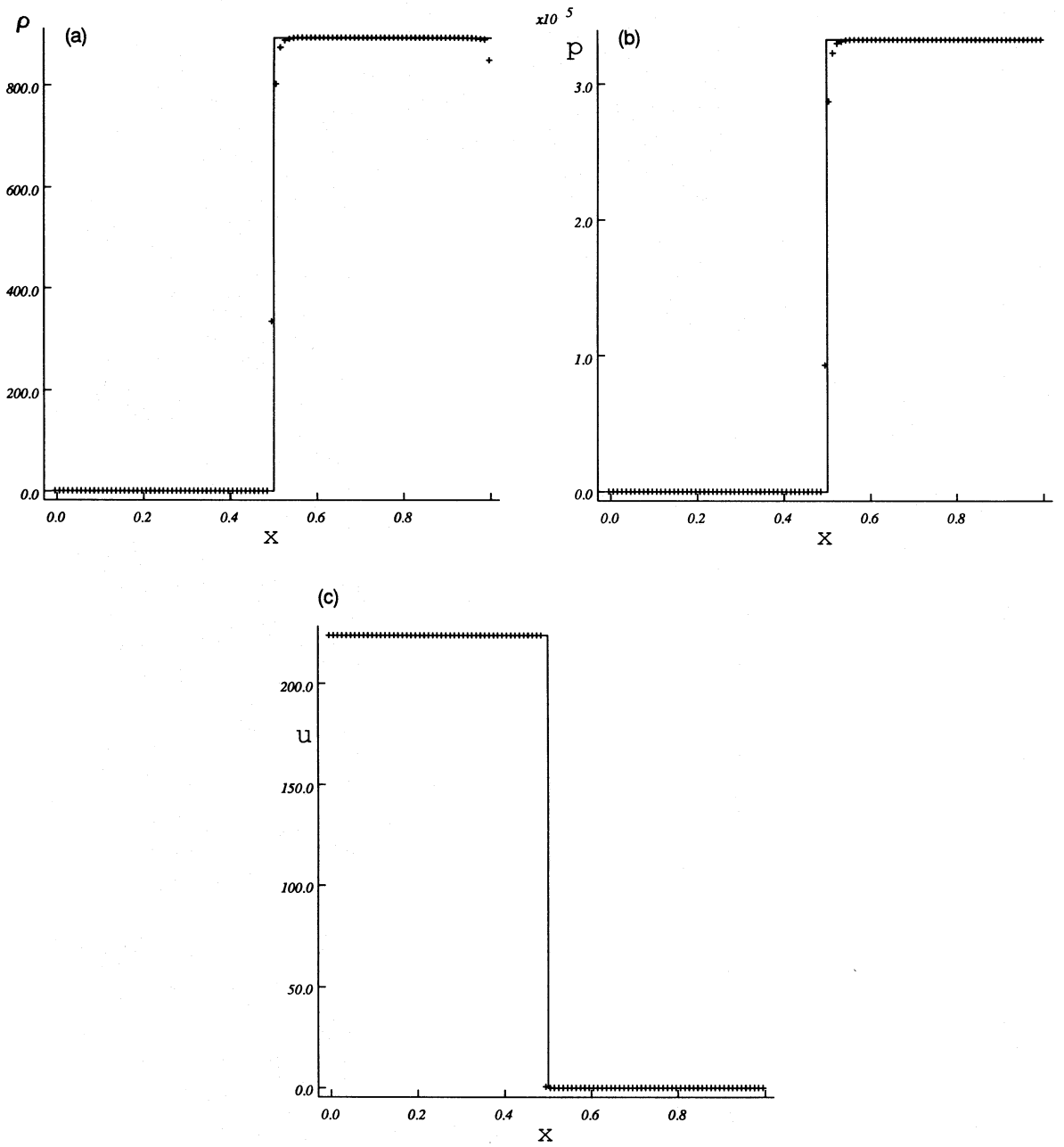


Figure 7. Exact solution (line) and numerical calculation (markers) for case 2 with $\gamma = 4/3$ at $t = 1.5$ (Riemann solver A): (a) density; (b) pressure; (c) x-component of 4-velocity.

gives us some confidence that the code is reliable in more than one dimension.

Case 4. We did not need artificial viscosity in any of the calculations we have described so far, since none of the two-dimensional calculations involve a strong shock propagating almost parallel to the grid. However, such shocks do occur in jet calculations, and they tend to suffer from the kind of distortion described by Quirk (1992). Fig. 9 shows a jet calculation both with and without an artificial viscosity of the form given in Section 3. In the calculation without artificial viscosity, there are small bulges in the jet and bow shocks, and a large one in the contact discontinuity. These disappear if we set the artificial viscosity parameter, η, in equation (3.22) equal to 0.2. This amount of artificial viscosity has a negligible effect on the other calculations.

5 CONCLUSIONS

Although the relativistic equations are more complicated than those for classical gas dynamics, they are both strictly hyperbolic in the sense that the wavespeeds change monotonically across simple waves i.e. there are no compound waves such as in magnetohydrodynamics (Brio & Wu 1988). The solution to the Riemann problem is therefore unique, and one would expect modern upwind schemes to be just as

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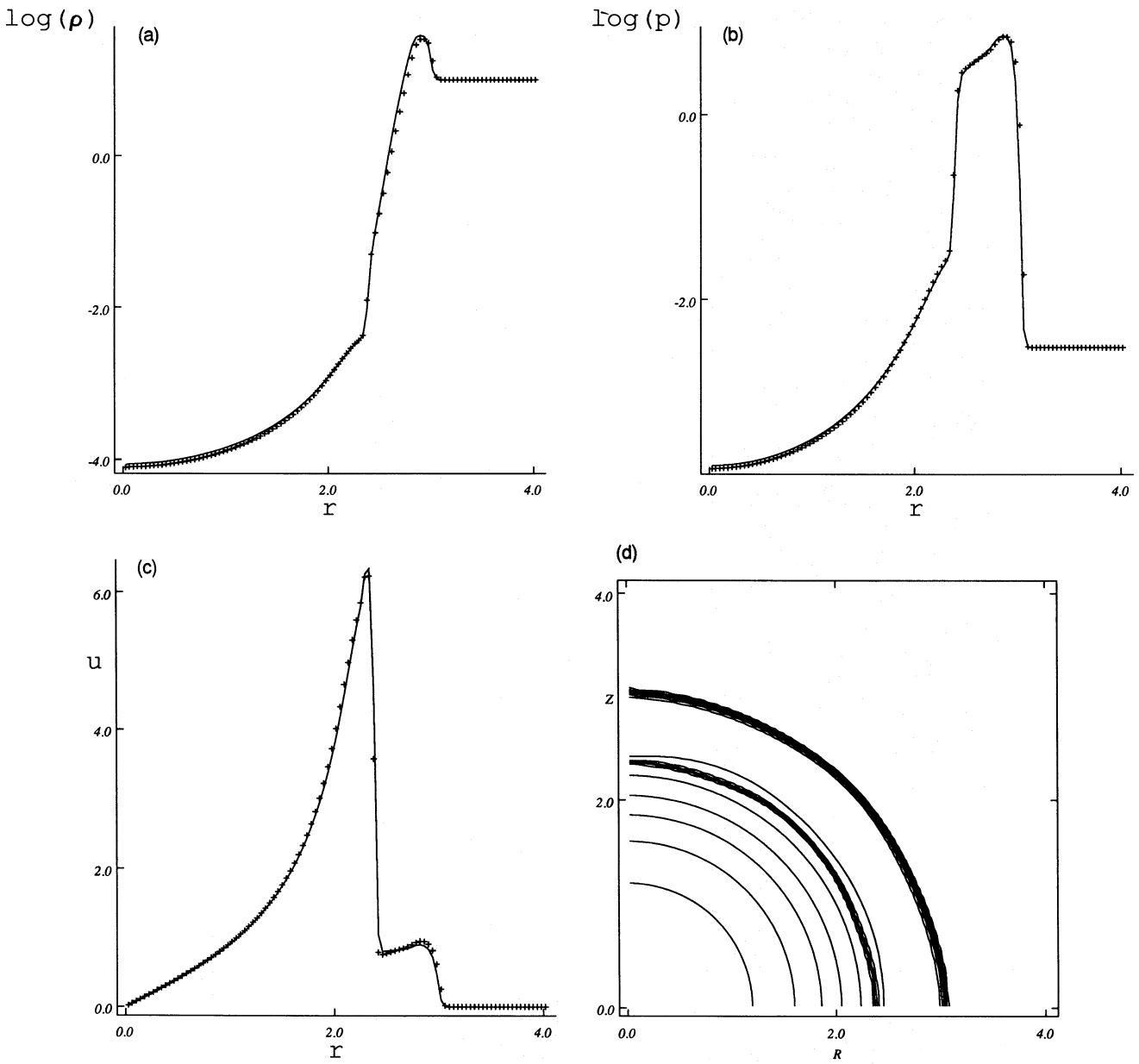


Figure 8. A cut at $z = 0$ showing the spherical (line) and axisymmetric (markers) numerical solutions at $t = 3$ for case 3 using Riemann solver A and the equation of state for an electron–proton gas: (a) density; (b) pressure; (c) r-component of 4-velocity; (d) logarithmic contours of pressure for the axisymmetric calculation.

effective in this case as they are in classical gas dynamics. Our results, as well as those of other workers, show that this is indeed true, but it is very expensive to solve the relativistic Riemann problem exactly without using a simplified equation of state. Unfortunately, this is a poor approximation for most astrophysical flows.

It is therefore gratifying that an approximate Riemann solver can cope with a realistic equation of state, even in extreme conditions. Our test cases, while not exhaustive, are more diverse than those considered by other authors, and fairly representative of the kinds of relativistic flow that can occur in astrophysics. The key ingredients seem to be the use of left and right Jacobians, and the exact treatment of strong rarefactions, instead of relying entirely on some mean constructed from the left and right states. Although we have only used this Riemann solver in one particular second-order scheme, there is no reason why it cannot be combined with some other upwind scheme, such as PPM.

The most obvious qualitative difference between the relativistic and classical case is that there is no limit to the compression across a relativistic shock. Such shocks therefore behave like radiative shocks, in the sense that they can generate thin dense shells in the post-shock gas, which require much higher numerical resolution than can easily be achieved on a uniform grid in more than one dimension. The only viable solution for this difficulty is to use an adaptive grid of the kind discussed in Section 3.

In this paper we have ignored processes, such as annihila-

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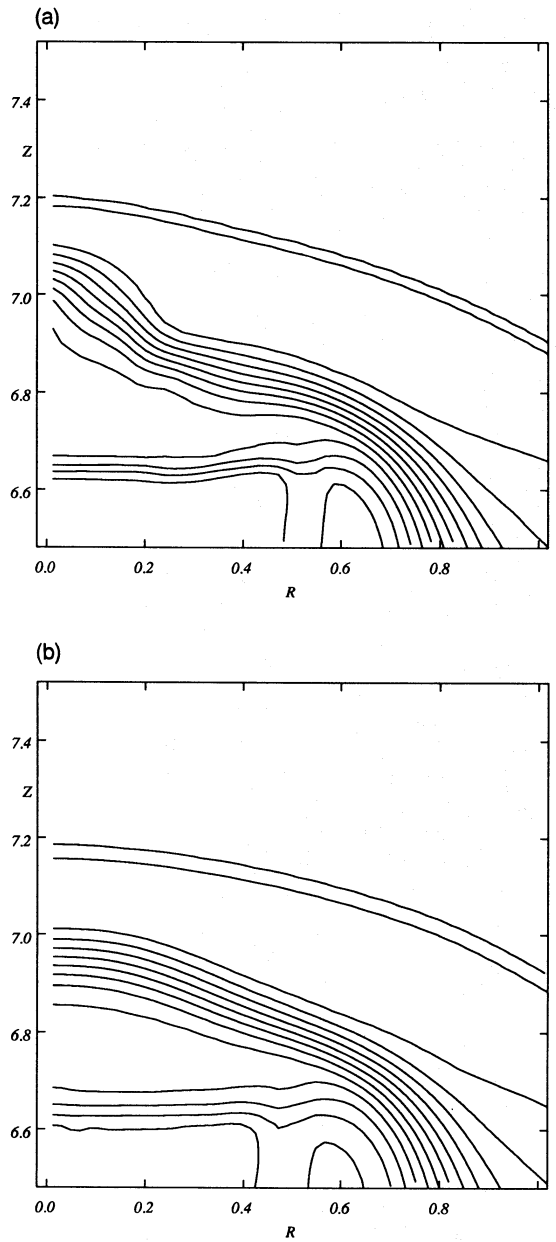


Figure 9. The working surface of a relativistic jet (case 4) at $t = 9$ using Riemann solver A and the equation of state for an electron–proton gas. 10 evenly spaced logarithmic contours of density between $\log \rho = -1.59$ and 1.63: (a) no artificial viscosity; (b) artificial viscosity parameter $\mu = 0.2$.

tion and radiative losses, that destroy conservation. However, one can deal with these by including appropriate source terms in the particle and energy conservation equations. Such terms do not cause any numerical difficulties unless they introduce very small scales into the problem. Even if they do, the adaptive grid should be able to cope with all but the most extreme cases.

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APPENDIX A

The Riemann solvers described in Section 3 require the eigenvalues and left and right eigenvectors of the matrices **A**, **B** in equations (3.7) and (3.13). It is slightly more convenient to start with equation (3.13) which we write as

$$\frac{\partial \boldsymbol{Q}}{\partial t} + \mathbf{B}_x \frac{\partial \boldsymbol{Q}}{\partial x} = 0,$$

where the suffix, x, denotes the x-direction and $\boldsymbol{Q}$ is given by equation (3.12)

$$\boldsymbol{Q} = [p, \Gamma v_x, \Gamma v_y, \Gamma v_z, \rho, \boldsymbol{c}]^{\mathrm{T}} = [p, u^x, u^y, u^z, \rho, \boldsymbol{c}]^{\mathrm{T}}.$$

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Then $\mathbf{B}_x$ is of the form

$$\mathbf{B}_x = \begin{pmatrix} \mathbf{B}_x^1 & 0 \\ 0 & \mathbf{B}_x^2 \end{pmatrix}$$

where $\mathbf{B}_x^1$ is the 5×5 matrix

$$\mathbf{B}_x^1 = \begin{pmatrix} \frac{\Gamma u^x(1-a^2)}{\Delta} & \frac{wa^2\Gamma}{\Delta\Gamma_x^2} & -\frac{wa^2 v_x u^y}{\Delta} & -\frac{wa^2 v_x u^z}{\Delta} & 0 \\ \frac{u^x v_x a^2}{w\Delta}+\frac{1}{\Gamma w} & v_x-\frac{u^x a^2\Gamma}{\Delta\Gamma_x^2} & \frac{u^x u^y v_x a^2}{\Delta} & \frac{u^x u^z v_x a^2}{\Delta} & 0 \\ \frac{u^y v_x a^2}{w\Delta} & -\frac{u^y a^2\Gamma}{\Delta\Gamma_x^2} & v_x+\frac{u^y u^y v_x a^2}{\Delta} & \frac{u^y u^z v_x a^2}{\Delta} & 0 \\ \frac{u^z v_x a^2}{w\Delta} & -\frac{u^z a^2\Gamma}{\Delta\Gamma_x^2} & \frac{u^y u^z v_x a^2}{\Delta} & v_x+\frac{u^z u^z v_x a^2}{\Delta} & 0 \\ -\frac{v_x\rho}{w\Delta} & \frac{\Gamma\rho}{\Delta\Gamma_x^2} & -\frac{u^y v_x\rho}{\Delta} & -\frac{u^z v_x\rho}{\Delta} & v_x \end{pmatrix},$$

where

$$\Delta = \Gamma^2(1-a^2)+a^2$$

and

$$\Gamma_x = (1-v_x^2)^{-1/2}.$$

$\mathbf{B}_x^2$ describes advection of the different particle species and is simply $\mathbf{B}_x^2 = v_x I$, where I is the $(N-1)\times(N-1)$ unit matrix.

The eigenvalues of $\mathbf{B}_x^1$ corresponding to sound waves are

$$\lambda_x^{\pm} = v_x - \left(\frac{v_x a^2 \mp a[\Gamma^2(1-a^2)\Gamma_x^{-2}+a^2]^{1/2}}{\Delta}\right) \tag{A1}$$

and their associated right and left eigenvectors are

$$\boldsymbol{R}_x^{\pm} = \left[wa^2, -v_x\Gamma a^2 - \frac{a^2}{\Gamma}(v_x-\lambda_x^{\pm})^{-1}, -v_y\Gamma a^2, -v_z\Gamma a^2, \rho\right]^{\mathrm{T}},$$

$$\boldsymbol{L}_x^{\pm} = \frac{1}{2\alpha^{\pm}}\left[\frac{\alpha^{\pm}}{wa^2}, -\Gamma\Gamma_x^{-2}, \Gamma v_x v_y, \Gamma v_x v_z, 0\right],$$

$$\alpha^{\pm} = [(v_x-\lambda_x^{\pm})\Delta - v_x a^2].$$

The other eigenvectors all correspond to the eigenvalue, v_x, associated with contact discontinuity. We do not really need these for either Riemann solver, since the resolved state can be determined by considering only the left sound wave if $v_x > 0$, and the right one otherwise.

The eigenvalues for the y- and z-directions can be obtained by replacing v_x in equation (A1) by v_y and v_z respectively. The eigenvectors for the y-direction are

$$\boldsymbol{R}_y^{\pm} = \left[wa^2, -v_x\Gamma a^2, -v_y\Gamma a^2 - \frac{a^2}{\Gamma}(v_y-\lambda_y^{\pm})^{-1}, -v_z\Gamma a^2, \rho\right]^{\mathrm{T}},$$

$$\boldsymbol{L}_y^{\pm} = \frac{1}{2\alpha^{\pm}}\left[\frac{\alpha^{\pm}}{wa^2}, \Gamma v_x v_y, -\Gamma\Gamma_y^{-2}, \Gamma v_y v_z, 0\right],$$

$$\alpha^{\pm} = [(v_y-\lambda_y^{\pm})\Delta - v_y a^2].$$

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while those for the z-direction are

$$\boldsymbol{R}_z^{\pm}=\left[wa^2,\ -v_x\Gamma a^2,\ -v_y\Gamma a^2,\ -v_z\Gamma a^2-\frac{a^2}{\Gamma}(v_z-\lambda_z^{\pm})^{-1},\ \rho\right]^{\mathrm{T}},$$

$$\boldsymbol{L}_z^{\pm}=\frac{1}{2\alpha^{\pm}}\left[\frac{\alpha^{\pm}}{wa^2},\ \Gamma v_x v_z,\ \Gamma v_y v_z,\ -\Gamma\Gamma_z^{-2},\ 0\right],$$

$$\alpha^{\pm}=[(v_z-\lambda_z^{\pm})\Delta-v_z a^2].$$

The variables defined by equation (3.6)

$$\boldsymbol{P}=[p, v_x, v_y, v_z, \rho, \boldsymbol{c}]^{\mathrm{T}}$$

satisfy

$$\frac{\partial \boldsymbol{P}}{\partial t}+\mathbf{A}_x\frac{\partial \boldsymbol{P}}{\partial x}=0.$$

The matrix $\mathbf{A}_x$ can be partitioned in the same way as $\mathbf{B}_x$

$$\mathbf{A}_x=\begin{pmatrix}\mathbf{A}_x^1 & 0\\ 0 & \mathbf{A}_x^2\end{pmatrix}$$

where $\mathbf{A}_x$ is a 5×5 matrix and $\mathbf{A}_x^2=\mathbf{B}_x^2$. The eigenvectors of $\mathbf{A}_x^1$ can most easily be determined from those of $\mathbf{B}_x^1$ by using the transformation between the $\boldsymbol{P}$ and $\boldsymbol{Q}$ variables. If we split these into two parts,

$$\boldsymbol{P}=[\boldsymbol{P}^1, \boldsymbol{P}^2],\qquad \boldsymbol{P}^1=[p, v_x, v_y, v_z, \rho]^{\mathrm{T}},\qquad \boldsymbol{P}^2=[\boldsymbol{c}]^{\mathrm{T}},$$

$$\boldsymbol{Q}=[\boldsymbol{Q}^1, \boldsymbol{Q}^2],\qquad \boldsymbol{Q}^1=[p, u_x, u_y, u_z, \rho]^{\mathrm{T}},\qquad \boldsymbol{Q}^2=\boldsymbol{P}^2,$$

then

$$\mathrm{d}\boldsymbol{P}^1=C\,\mathrm{d}\boldsymbol{Q}^1,$$

where

$$\mathbf{C}=\begin{pmatrix}1 & 0 & 0 & 0 & 0\\ 0 & \dfrac{1}{\Gamma\Gamma_x^2} & -\dfrac{v_xv_y}{\Gamma} & -\dfrac{v_xv_z}{\Gamma} & 0\\ 0 & -\dfrac{v_xv_y}{\Gamma} & \dfrac{1}{\Gamma\Gamma_y^2} & -\dfrac{v_yv_z}{\Gamma} & 0\\ 0 & -\dfrac{v_xv_z}{\Gamma} & -\dfrac{v_yv_z}{\Gamma} & \dfrac{1}{\Gamma\Gamma_z^2} & 0\\ 0 & 0 & 0 & 0 & 1\end{pmatrix}$$

and

$$\mathbf{C}^{-1}=\begin{pmatrix}1 & 0 & 0 & 0 & 0\\ 0 & \Gamma+\Gamma^3v_x^2 & \Gamma^3v_xv_y & \Gamma^3v_xv_z & 0\\ 0 & \Gamma^3v_yv_x & \Gamma+\Gamma^3v_y^2 & \Gamma^3v_yv_z & 0\\ 0 & \Gamma^3v_zv_x & \Gamma^3v_yv_z & \Gamma+\Gamma^3v_z^2 & 0\\ 0 & 0 & 0 & 0 & 1\end{pmatrix}.$$

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Then if $\boldsymbol{R}_{\mathbf{A}}$, $\boldsymbol{L}_{\mathbf{A}}$ and $\boldsymbol{R}_{\mathbf{B}}$, $\boldsymbol{L}_{\mathbf{B}}$ are left, right eigenvectors of $\mathbf{A}_x^1$ and $\mathbf{B}_x^1$ corresponding to the same eigenvalue, we have

$$\boldsymbol{R}_{\mathbf{A}} = C\boldsymbol{R}_{\mathbf{B}}, \qquad \boldsymbol{L}_{\mathbf{A}} = \boldsymbol{L}_{\mathbf{B}}\mathbf{C}^{-1}.$$

The sound wave eigenvectors for matrix $\mathbf{A}^1$ are then

$$\boldsymbol{R}_x^{\pm} = \left[wa^2, -\frac{\alpha^{\pm}}{\Gamma^2}, \frac{a^2\lambda_x^{\pm}v_y}{\Gamma(v_x-\lambda_x^{\pm})}, \frac{a^2\lambda_x^{\pm}v_z}{\Gamma(v_x-\lambda_x^{\pm})}, \rho\right]^{\mathrm{T}},$$

$$\boldsymbol{L}_x^{\pm} = \frac{1}{2\alpha^{\pm}}\left[\frac{\alpha^{\pm}}{wa^2}, -\Gamma^2, 0, 0, 0\right],$$

$$\alpha^{\pm} = [(v_x-\lambda_x^{\pm})\Delta - v_x a^2],$$

for the x-direction,

$$\boldsymbol{R}_y^{\pm} = \left[wa^2, \frac{a^2\lambda_y^{\pm}v_x}{\Gamma(v_y-\lambda_y^{\pm})}, -\frac{\alpha^{\pm}}{\Gamma^2}, \frac{a^2\lambda_y^{\pm}v_z}{\Gamma(v_y-\lambda_y^{\pm})}, \rho\right]^{\mathrm{T}},$$

$$\boldsymbol{L}_y^{\pm} = \frac{1}{2\alpha^{\pm}}\left[\frac{\alpha^{\pm}}{wa^2}, 0, -\Gamma^2, 0, 0\right],$$

$$\alpha^{\pm} = [(v_y-\lambda_y^{\pm})\Delta - v_y a^2],$$

for the y-direction and

$$\boldsymbol{R}_z^{\pm} = \left[wa^2, \frac{a^2\lambda_z^{\pm}v_x}{\Gamma(v_z-\lambda_z^{\pm})}, \frac{a^2\lambda_z^{\pm}v_y}{\Gamma(v_z-\lambda_z^{\pm})}, -\frac{\alpha^{\pm}}{\Gamma^2}, \rho\right]^{\mathrm{T}},$$

$$\boldsymbol{L}_x^{\pm} = \frac{1}{2\alpha^{\pm}}\left[\frac{\alpha^{\pm}}{wa^2}, 0, 0, -\Gamma^2, 0\right],$$

$$\alpha^{\pm} = [(v_z-\lambda_z^{\pm})\Delta - v_z a^2],$$

for the z-direction.

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